

**MEDIMASYST: A DEVELOPMENT OF MEDICINE
INVENTORY MANAGEMENT SYSTEM FOR
ARGUELLES PHARMACY**



**A System Analysis and Design Presented to the
Faculty of the Institute of Computing
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ABSTRACT

Arguelles Pharmacy, established by Dr. Gemma Lourdes Arguelles Maligad in 2003, is a thriving business with five branches, each delivering dedicated service and efficient operations. Despite its success, the pharmacy faces challenges in inventory management, including overstocking, understocking, and delays in pharmaceutical supply deliveries. The reliance on conventional paper-based systems exacerbates these issues, affecting operational efficiency and customer satisfaction.

This study aims to develop a robust Medicine Inventory Management System (MIMS) to address these challenges. By leveraging technology, automation, and data analytics, the proposed system seeks to streamline inventory management processes, enhance accuracy, and ensure uninterrupted medication availability. The system will allow the admin to manage customer details, orders, and stock levels effectively, generate receipts, and produce detailed sales and inventory reports.

The methodology employs the Modified Waterfall Model, ensuring systematic development and risk mitigation. The project involves a comprehensive analysis of the pharmacy's existing processes, identifying inefficiencies and proposing new approaches for system improvement. Various diagrams, including use case, activity, and data flow diagrams, illustrate the interactions and data flow within the system.

The implementation of the MIMS will significantly enhance operational efficiency, reduce manual errors, and improve overall service quality at Arguelles Pharmacy. By addressing the key issues of stock management and delivery delays, the system will contribute to better customer satisfaction and sustained business growth. The study highlights the importance of effective inventory control in the healthcare sector, emphasizing patient safety, cost control, and operational efficiency.

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TABLE OF CONTENTS

TITLE PAGE.....	i
ABSTRACT	ii
ACKNOWLEDGEMENT	iii
TABLE OF CONTENTS	iv
LIST OF TABLES	vi
LIST OF FIGURES.....	vi
CHAPTER I	1
INTRODUCTION.....	1
A. Company Overview	1
B. Organizational Structure	2
C. Problems Encountered	3
D. Existing Business Process	3
E. Objectives Of The Project.....	6
General Objectives.....	6
Specific Objectives.....	6
F. Scope And Limitations	7
G. Related Systems And Works	10
CHAPTER II	12
METHODOLOGY	12
A. Project Development Methodology.....	12

Work Breakdown Structure	13
Gantt Chart	14
B. System Analysis	15
Process Model: Use Case Diagram and Description	17
Process Model: Activity Diagram.....	21
Data Model: Context Data Flow Diagram.....	23
Data Model: Level 1 Data Flow Diagram.....	23
Database Design: Entity-Relationship Diagram	25
CHAPTER III	26
RESULTS AND DISCUSSION	26
A. System Design	26
Data Dictionary.....	26
User Interface Design	30
CHAPTER IV.....	34
CONCLUSION AND RECOMMENDATION	34
A. Conclusion	34
B. Recommendation	35
REFERENCES.....	1
APPENDICES	3
Letter to the Organization	3
Environmental Scanning Documentation.....	4

LIST OF TABLES

Table 1. Scope and Limitations of the Project.....	8
Table 2. Use Case Description for [use case name]	17

LIST OF FIGURES

Figure 1. Organizational Structure.....	2
Figure 2. Existing Business Process	4
Figure 3. Supplier to Pharmacy Transaction Process.....	5
Figure 4. Modified Waterfall Life Cycle Model	12
Figure 5. Work Breakdown Structure of the Project.....	13
Figure 6. Gantt Chart of the Project.....	14
Figure 7. Use Case Diagram	17
Figure 8. Activity Diagram for Manage Order and Generate Receipt	21
Figure 9. Activity Diagram for Managing Stocks and Generate Reports.....	22
Figure 10. Context Data Flow Diagram	23
Figure 11. Level 1 Data Flow Diagram	24
Figure 12. Entity-Relationship Diagram.....	25
Figure 13. Log-in	30
Figure 14. Dashboard.....	31
Figure 15. Product.....	31
Figure 16. Order Details	32
Figure 17. Add Medicine	32
Figure 18. Sales Report.....	33
Figure 19. Inventory Report.....	33

CHAPTER I

INTRODUCTION

A. Company Overview

Arguelles Pharmacy embarked on its journey rooted in the dynamic interaction between its founder, Dr. Gemma Lourdes Arguelles Maligad, and medicine distributors. Facing financial challenges, Dr. Gemma engaged in a strategic agreement with a distributor, borrowing both cash and medicines to sustain her struggling business. An unexpected twist occurred when the distributor proposed settling the debt by accepting payment in the form of medicines rather than cash, a proposition agreed upon by both parties. As the payment deadline approached, the business faced further losses, leading to the decision to repay the distributor with medicines instead of the originally agreed-upon cash.

In response to these challenges, Dr. Gemma Lourdes Arguelles Maligad pondered a new direction for her enterprise. The idea emerged to establish a pharmacy, putting her in control of the medicine supply. In 2003, Arguelles Pharmacy was born, and over time, it flourished with the establishment of five branches. Each branch is managed by dedicated Admin/Purchaser, ensuring personalized service and efficient operation.

Located at Somoso Anita Bldg., Purok Mangga, Brgy. Sto Niño, Arguelles Pharmacy thrives on the unique journey that initiated its existence. The business model, forged from the innovative interaction between the owner and distributors, highlights adaptability and resilience. Driven by a commitment to excellence, the pharmacy optimizes inventory management through technology, automation, and data analytics.

Arguelles Pharmacy's success story is a testament to Dr. Gemma Lourdes Arguelles Maligad's entrepreneurial spirit. From navigating financial challenges to establishing a thriving pharmacy, the company showcases resilience, adaptability, and a dedication to meeting community healthcare needs. The five branches

symbolize a commitment to accessibility, quality service, and community engagement, fostering a sense of ownership in each branch's management team.

The four staff members at the mentioned branch contribute significantly to the pharmacy's success. Their roles encompass various aspects of customer service, medicine dispensing, and consultation assistance. Their dedication ensures that the community's healthcare needs are met with efficiency and care.

With an estimated daily footfall of 250-plus individuals per branch, Arguelles Pharmacy serves over 1,750 people per week and exceeds 50,000 individuals per month. This impressive customer base not only includes those purchasing medicines but also individuals seeking consultations and inquiring about medications. The widespread popularity of Arguelles Pharmacy is a testament to its reputation for quality service, accessibility, and community engagement.

B. Organizational Structure

Figure 1 illustrates the Arguelles Pharmacy setup, with an Admin overseeing finances and branch management. The four pharmacy staff members are crucial in assisting customers and offering advice on treatments, collaborating to ensure the pharmacy operates smoothly and achieves success.

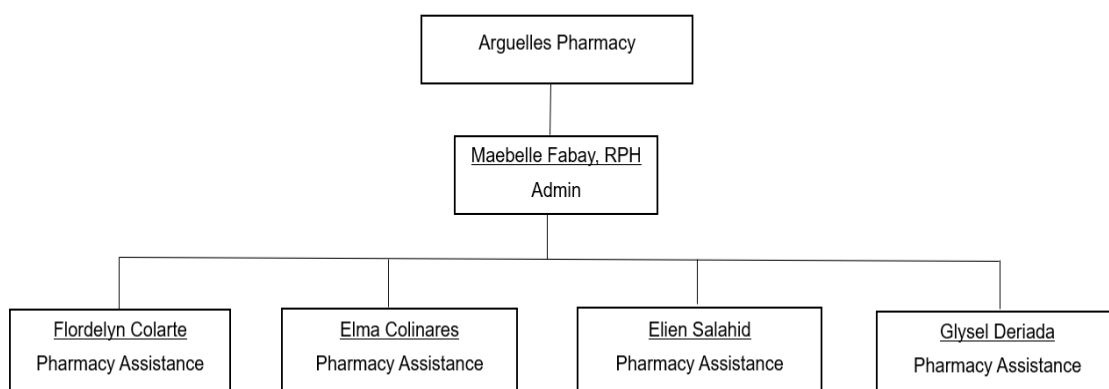


Figure 1. Organizational Structure

C. Problems Encountered

Arguelles Pharmacy faces challenges in effectively managing its inventory, leading to issues such as overstocking, understocking, and outdated product management. Conventional paper-based systems and limited point-of-sale solutions contribute to inefficiencies. The pharmacy also grapples with delays in pharmaceutical supply delivery, relying on independent distributors, introducing variability in quality, pricing, and delivery schedules. These challenges hinder the pharmacy's goal of ensuring uninterrupted medication availability. Addressing these issues is crucial for enhancing overall efficiency and providing reliable pharmaceutical services to customers.

The problems encountered by Arguelles Pharmacy are:

- **Out of stock of medicine.** Arguelles Pharmacy runs out of important medicines, it hurts their earnings and makes customers unhappy. Customers might lose trust in the pharmacy and negative reviews. They might even go to other pharmacies to get the medicines they need. This shortage can also affect the health of customers who rely on these medications.
- **Delayed deliveries of medicine.** Whenever medicines arrive late at Arguelles Pharmacy from suppliers, the delays are frequently attributed to the reliance on independent distributors for the supply of medicines. Challenges arise when these distributors encounter issues such as inadequate stock, transportation challenges, or disruptions in the supply chain.
- **Manual recording processes in place.** Arguelles Pharmacy faces challenges due to the reliance on manual recording processes for inventory management. This method is prone to errors, making it difficult to accurately track stock levels and reorder medications.

D. Existing Business Process

The process begins with the customer's arrival. The first decision point checks if a prescription is available. If yes, the process proceeds to Submit Prescription, followed by Verification Successful. After verification, the standard procedure of payment, receiving medicine, optional counseling, and concluding

with Record Keeping is followed. If the customer doesn't have a prescription, the flow goes to Choose Medicine. Subsequently, after selecting the medicine, it proceeds to Verification Successful. Following this, the process continues with the standard payment, receiving medicine, optional counseling, and concluding with Record Keeping.

Figure 2 illustrates the customer-to-pharmacy transaction process, outlining the series of steps a customer undergoes when purchasing medicine at a pharmacy.

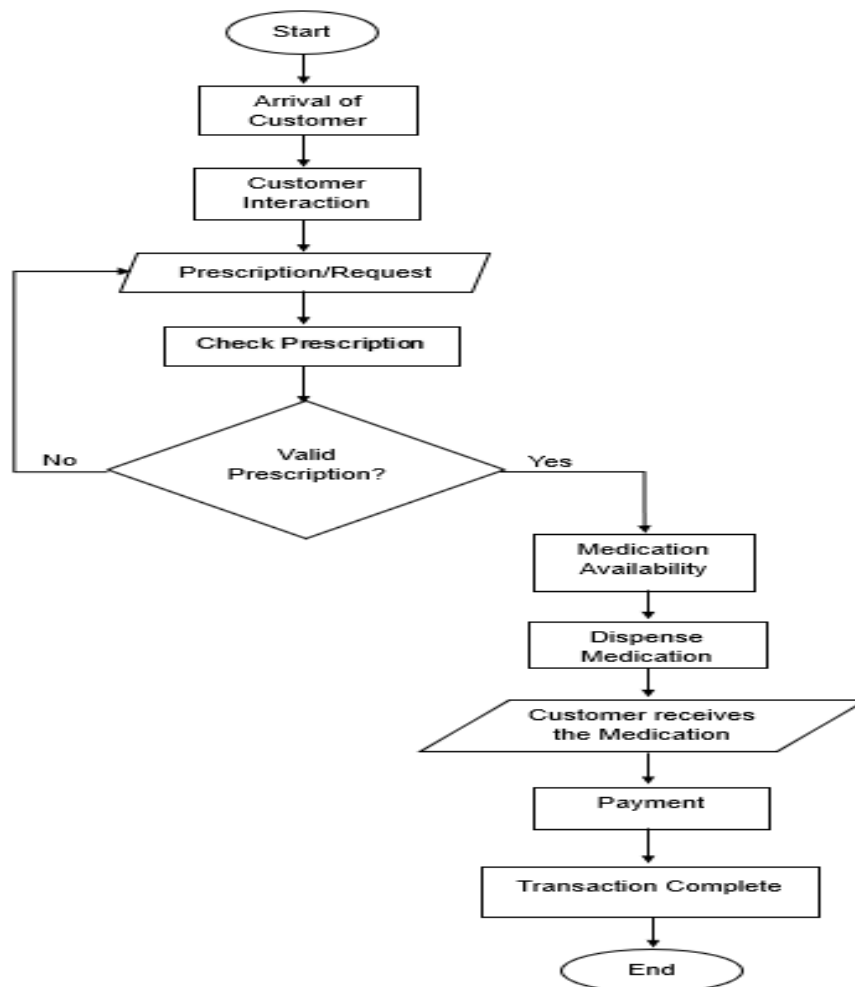


Figure 2. Existing Business Process

Arguelles Pharmacy follows a well-structured process for ordering out-of-stock medicines. The pharmacy relies on consistent suppliers to replenish best-selling items promptly. After identifying the specific medicines needed, including type and quality, a careful review of the paper-based inventory ensures accurate ordering. If the decision to order is affirmative, the pharmacy places the order, and upon receiving the medicines, the admin verifies the order's accuracy, especially the quantity. Financial transactions are then initiated, and post-transaction, the admin updates the inventory records for operational efficiency.

Figure 3 illustrates the journey of medicines from the supplier to the pharmacy business process, emphasizing the smooth integration into a traditional paper-based inventory system. This seamless process aligns with the maintenance of a feedback loop for continuous improvement.

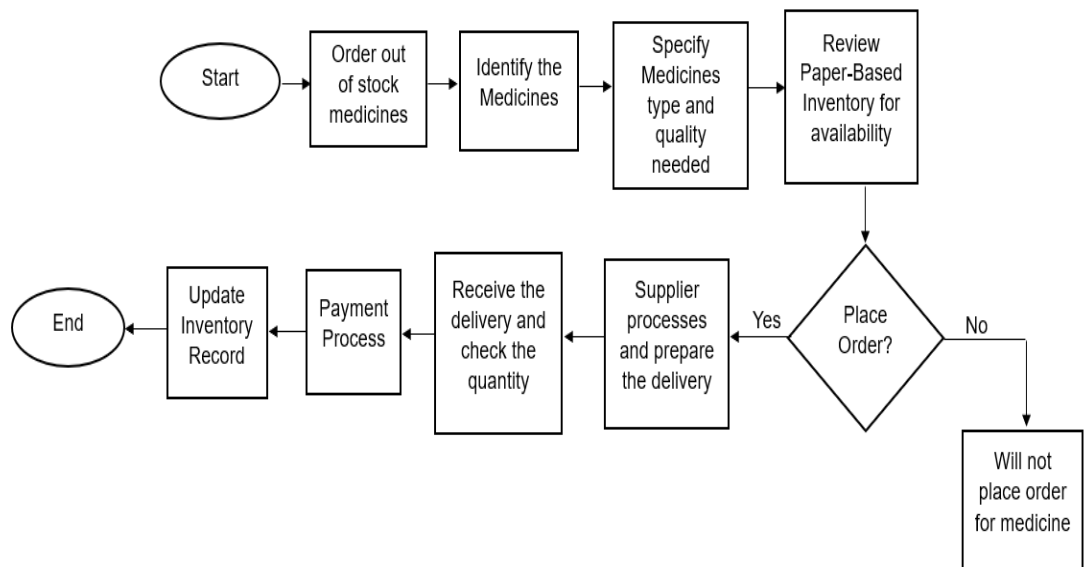


Figure 3. Supplier to Pharmacy Transaction Process

E. Objectives Of The Project

General Objectives

To enhance operational efficiency, this study aims to develop a robust Medicine Inventory Management System at Arguelles Pharmacy, with a primary focus on mitigating and preventing instances of medicine-related issues. Additionally, it seeks to accurately record inventory for the improvement of current routines and processes

Specific Objectives

The system specifically enables the following:

1. Allows Admin to input the customer details;
 - 1.1. Customer name;
 - 1.2. Address;
 - 1.3. Phone;
 - 1.4. Email;
2. Allow Admin to Input order details;
 - 2.1. Order;
 - 2.2. Product name;
 - 2.3. Quantity;
 - 2.4. Unit price;
 - 2.5. Total price;
 - 2.6. PO number;
 - 2.7. Batch number;
 - 2.8. Total payment;
3. Allow admin to manage the stocks;
 - 3.1. Date;
 - 3.2. Time;
 - 3.3. Product name;
 - 3.4. Manufacturer;
 - 3.5. Expiration date;
 - 3.6. Batch number;

- 3.7. Manufacturer;
- 3.8. Expiration date;
- 3.9. Batch number;
- 3.10. Dosage;
- 3.11. Quantity;
- 3.12. Price;
- 3.13. Total amount;
- 3.14. Image url;
- 4. Allow admin to manage customer's orders;
- 5. Generate receipt;
- 6. Allow admin to generate reports;
 - 6.1. Sales Reports;
 - 6.2. Inventory Reports;

F. Scope And Limitations

The scope of this project involves analyzing, designing and develop a comprehensive Medicine Inventory System with a primary focus on tracking medication information. Additionally, the system aims to generate customizable reports to meet the specific needs of users. Furthermore, the system will include features such as real-time inventory management and alerts for medication refills.

The system may encounter challenges in integrating with various pharmacy management systems, ensuring data security and privacy compliance, and relying on accurate input of medication information. Moreover, limitations may arise in handling medication recalls, and careful consideration of user access permissions is essential to prevent unauthorized access to sensitive health data.

Table 1. Scope and Limitations of the Project

People	Admin • Ability to input medicine, customer details, and manage stocks using an inventory management system, while also assessing the system's capability to generate reports and receipts.
Technology	Hardware Laptop • The laptop-based medicine inventory management system provides users with the capability to monitor all transactions, view inventory levels, identify out-of-stock items, and generate receipts and reports. However, its effectiveness may be limited by factors such as hardware constraints, software compatibility issues, and the need for accurate data entry. Software: My SQL • The preferred database software is MYSQL, where tables store customer order details, supplier details, and medicine details data, including their respective attributes. These datasets are essential for assessing the efficiency and effectiveness of the medicine inventory management system. Visual Code Studio • Visual Studio Code (VS Code) is an ideal choice for developing a Medicine Inventory Management System. Its versatile IDE environment supports coding,

	debugging, version control, and extensions, facilitating efficient development and customization of the system.
Data	Products • The system provides a clear overview of medicine details such as expiry dates, batch numbers, names, manufacturers, dosages, and costs. This ensures patient safety and streamlines inventory management for efficient operations. Orders • The system shows customer order details like order number, medicines, and the quantity ordered. This ensures accurate order fulfillment, efficient inventory management, and patient safety when using medicines.
Process	Inventory Management • The inventory management system for medicine keeps track of medicine levels, identifies shortages and excess, and monitors expired medicines and their costs. This helps ensure smooth operations, preventing stockouts and overstocking, while also enabling easy identification of expired medicines and their associated expenses. Order Management • Supplier reliability, inventory accuracy, communication barriers, external factors, resource constraints, quality control, and technology limitations challenge the order management process. Arguelles Pharmacy's order management includes proactive supplier sourcing, demand-based ordering, and hands-on supplier involvement. Admin oversees tracking, inventory, and timely delivery. Despite

	challenges, proactive monitoring and clear communication optimize the process.
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G. Related Systems And Works

In healthcare systems, effective management of medicine inventory is crucial to prevent both stockouts and overstocking. Inadequate inventory systems can lead to increased expenses and reduced availability of medicines. Stockouts occur when certain items are unavailable to fulfill customer orders, while overstocking happens when the number of items exceeds customer demand [1].

In a medical warehouse in Kuwait, medicine inventory system needed improvement as there was often stock-out or overstock of some types of medicines. As there were changes in the demand pattern and the medicine availability from the supplier, modelling and simulation were suitable tools to establish the most appropriate inventory policy for medicine supply chain [2].

The government has launched a program to support people, especially those who are already having their maintenance medicine and who cannot afford or financially support their needs. The fast and efficient distribution of medicine to everyone who needs it is very important. Managing medication supply is a challenge for all pharmacies. An essential part of every successful pharmacy operation, as keeping control of inventory levels has a major impact on bottom-line costs and profit margins. In a highly competitive industry that has seen large increases in supply costs, it's vital to manage inventory as effectively as possible [3].

The importance of inventory control is to ensure that the drugs should be supply at right time in the drug store, avoid wastage and shortage and maintain optimal stock and to reduce investment. The priority in procurement and inventory management is patient safety, An effective procurement process ensures the availability of drugs and medications that are appropriate to the patient's circumstances, at recognizable standards of quality, good inventory control supports procurement and utilizes appropriate systems to track shipments and

inventory, and to forecast needs to ensure that patients continue to have reasonable access to pharmaceutical products for their health and well-being and all members in the pharmacy who have a role in medication procurement and inventory management must receive training with respect to their legal obligations and expected standards of practice are some of the principles in medication procurement and inventory management [4].

Based on the current process of the Barangay Health Center, the main objective of this study is to be able to develop a Web based application that will function as automated medical health tracking through prescriptions of medications and medicine delivery tracking. As a result of the growing population of persons with disabilities and senior citizens in the city of Makati City Philippines, there is a high demand for medical consultations and medication. The need for process improvement that will answer the health tracking and prescriptive medicine delivery monitoring system is needed [5]. In the Philippines, effective inventory management is crucial for successful cost control in businesses. Timely inventory objectives, guided by software programs, help align inventory needs with economic orders. Tracking inventory status and vendor information is essential. A reliable inventory tracking system accelerates inventory processes, beyond just cost reduction. Inventories include stored materials, in-process items, and finished goods across various sectors. Despite managerial concerns, inventories are vital assets, albeit with storage costs and risks of spoilage or obsolescence [6].

In conclusion, effective management of medicine inventory is essential in healthcare systems to avoid stockouts and overstocking, ensuring medicines are available when needed. Governments and pharmacies alike strive to efficiently distribute medicines, especially to those in need. Proper inventory control not only impacts costs and profit margins but also ensures patient safety and access to quality medications. Developing automated systems, like web-based applications, can improve medication tracking and delivery processes, addressing the increasing demand for healthcare services. In the Philippines, as in other

countries, reliable inventory management is crucial for businesses to control costs and streamline operations while meeting customer needs effectively.

CHAPTER II

METHODOLOGY

A. Project Development Methodology

The Modified Waterfall Model, a conventional approach to software development allows a return to a previous phase for verification or validation, ideally confined to connecting steps [7]. It is suitable for the project as it helps to organize work and mitigate risks. The purpose of this model in our project is to allow us to clearly define each stage of the process. Additionally, if there is an error, we can go back to the incorrect part.

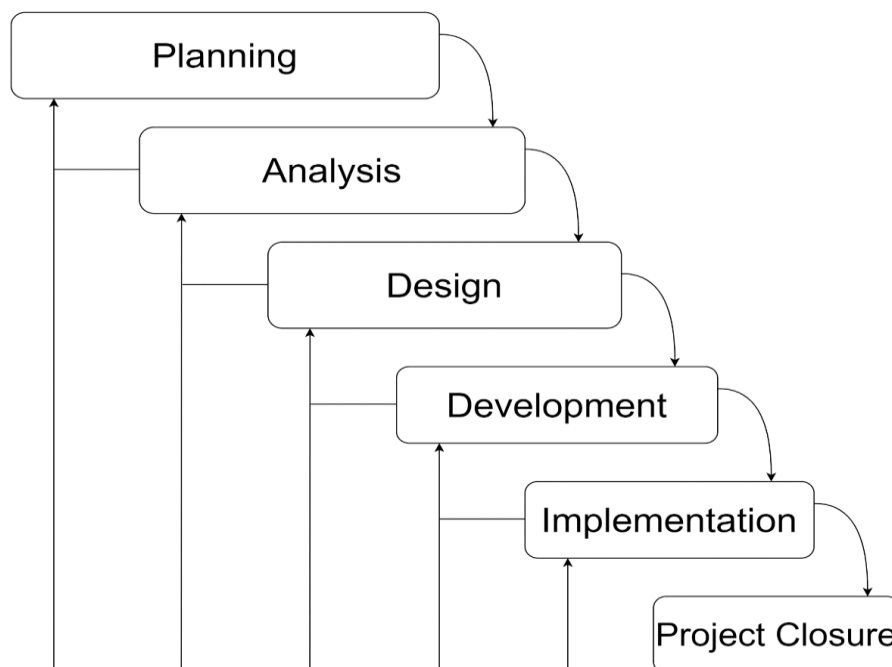


Figure 4. Modified Waterfall Life Cycle Model

Work Breakdown Structure

A work breakdown structure (WBS) is a project management tool that uses a step-by-step process to accomplish huge projects with many moving parts [8]. The process allows for correction in case of errors in a process, enabling adjustments to be made to the system that has been developed. By using a WBS, the team can focus on their specific tasks and deliverables while also seeing how their piece fits into the project as a whole [8].

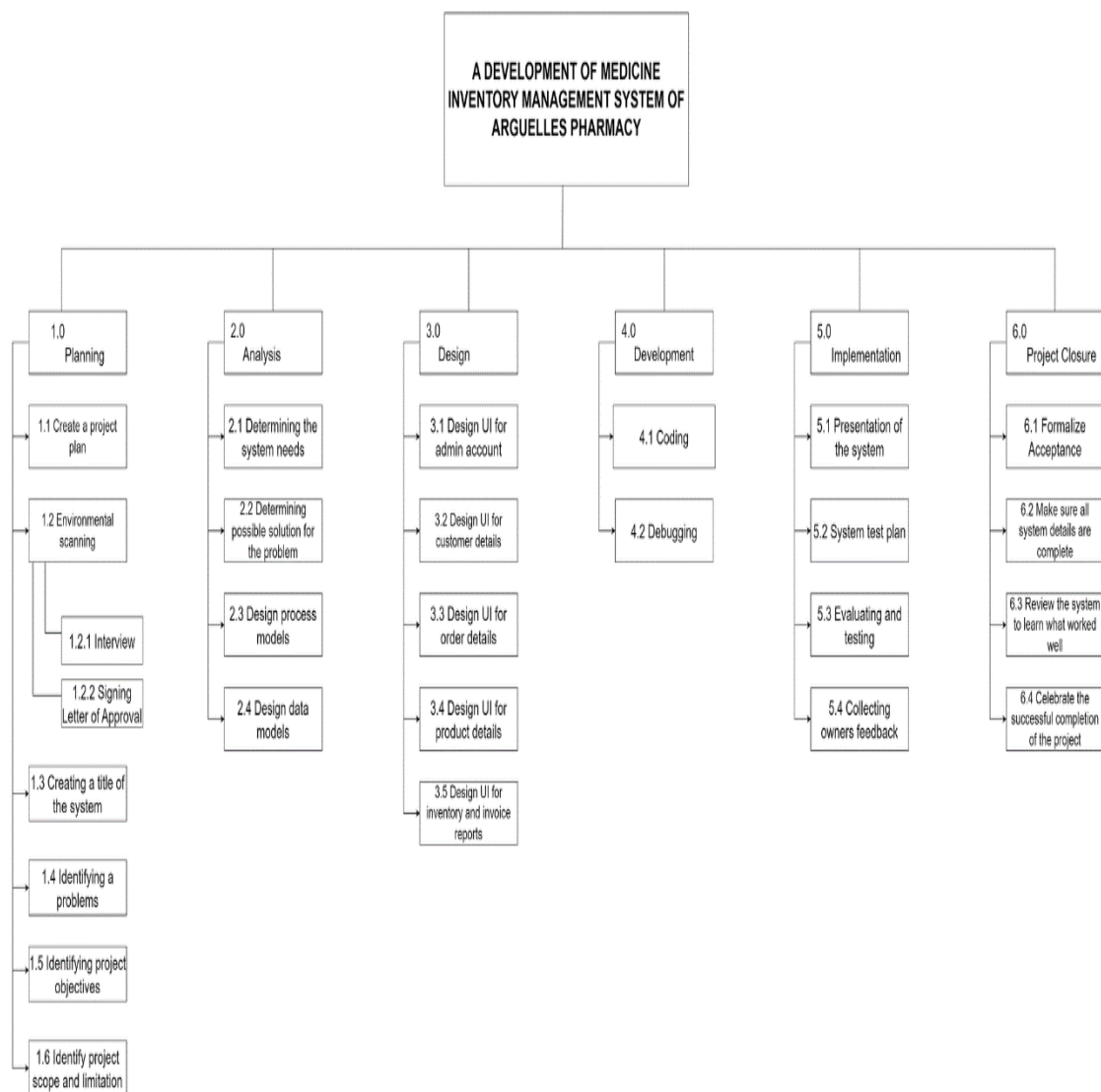


Figure 5. Work Breakdown Structure of the Project

Gantt Chart

The Gantt chart is the most commonly used in project management [9]. It helps in the scheduling, management, and monitoring of certain tasks and resources within a Medicine inventory management system project. The chart depicts the project timetable, which includes both scheduled and accomplished tasks across time.

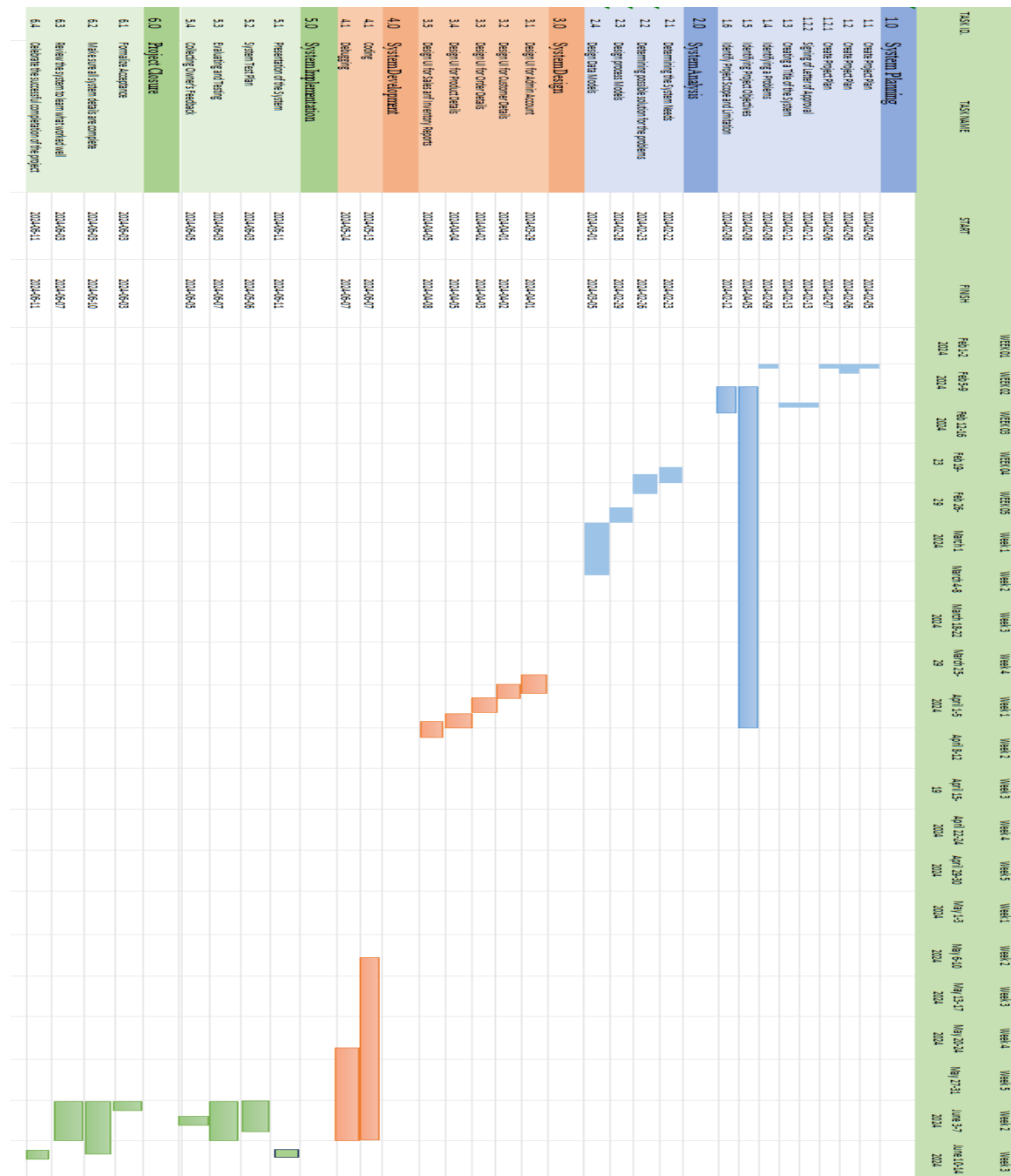


Figure 6. Gantt Chart of the Project

B. System Analysis

The project team explored different systems to gather ideas for improving the pharmacy's current setup. They came up with new approaches to simplify the system and enhance its functionality. To gain a better understanding of how the system works, they used various types of diagrams such as use case diagrams, activity diagrams, context data flow diagrams, and level 1 data flow diagrams. These diagrams helped visualize how information flows within the software and identify the different interactions between users and the system.

The Medicine Inventory Management System at Arguelles Pharmacy must efficiently handle functional requirements such as recording customer orders, managing medicine details, computing costs, and generating receipts. Additionally, it must meet non-functional requirements, including technical specifications, performance, usability, and security considerations, to support seamless pharmacy operations.

Functional Requirements

- Recording the Customer Details
- Recording the Order Details
- Recording the Product Details
- Compute the Total Cost of Medicine
- Generate Receipts

Non-Functional Requirements

Technical:

- The system must be accessible on the laptop hardware to connect medicine inventory management system to Arguelles Pharmacy.

Performance:

- The performance of the Medicine Inventory Management System at Arguelles Pharmacy must support smooth operations, ensuring quick responses, multitasking capabilities, and efficient data storage.

Usability:

- The medicine inventory management system at Arguelles Pharmacy focuses on intuitive design for easy navigation and task completion, enhancing the admin experience and operational efficiency.

Security:

- The medicine inventory management system at Arguelles Pharmacy includes strict access controls, encryption of network communication and data storage, secure authentication methods, regular updates, and a disaster recovery plan.

Process Model: Use Case Diagram and Description

This Figure 6 represents the use case diagram, which greatly assists in making the tasks of the supplier, customer, and admin efficient and effective in achieving their respective goals at Arguelles Pharmacy through their various activities.

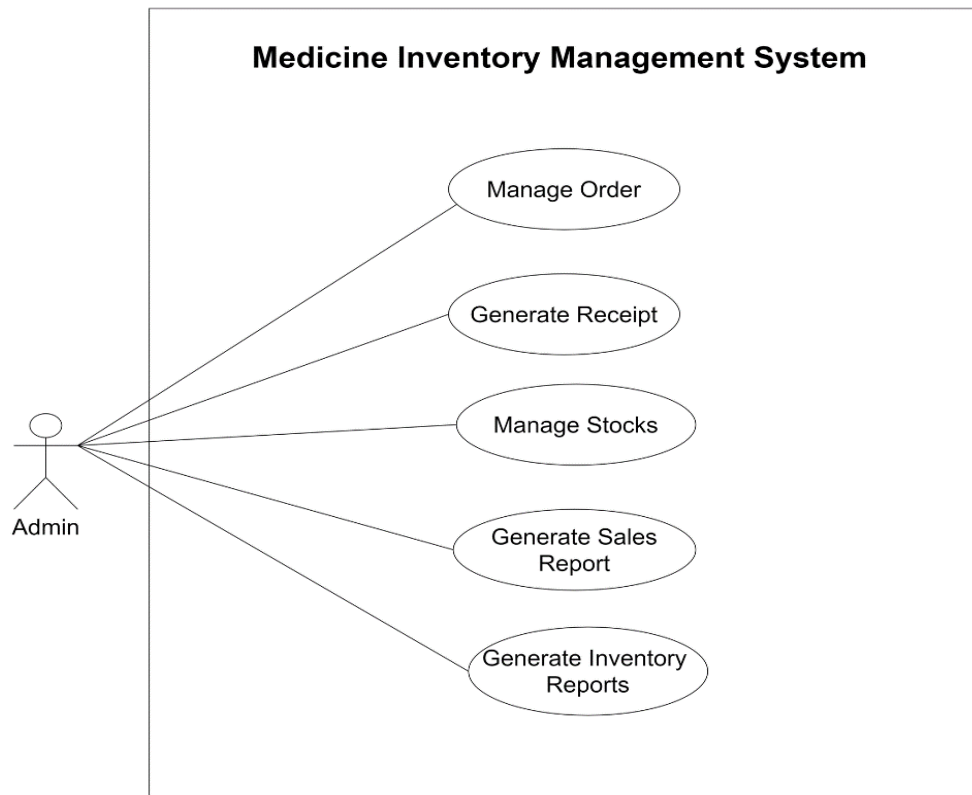


Figure 7. Use Case Diagram

The Table shows the use case "Manage Orders" and "Generate Receipt" involving the admin as the participating actor. The admin plays a key role in overseeing the entire order fulfillment process, from collecting customer orders to processing payments, updating inventories, and creating appropriate reports. The Admin's attention to detail and good management of the system are essential to the Medicine Inventory management System's success.

Table 2. Use Case Description for Manage Orders and Generate Receipt

Use Case Name	Manage Orders, Generate Receipt
Participating Actors	Admin

Entry Conditions	• The admin initiates the process within the system. • The admin manages customer orders and inventory.
Flow of Events	1. The Admin accepts customer orders. 2. The Admin inputs the customer orders into the system. 3. There is a decision point asking "Add Order?" If the answer is "Yes," the Admin goes back to input additional customer orders. If the answer is "No," the Admin proceeds to the next step. 4. The Admin accepts payment for the orders. 5. After the payment is accepted, the flow transitions to the System section. 6. The System generates a receipt for the transaction. 7. The process ends the System section.
Exit Conditions	• The admin oversees the process and updates the inventory accordingly.
Quality Requirements	• The admin ensures accurate management of orders,

	payments, and inventory reports within the system.
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The admin plays an essential role in managing the stock within the system. Responsible for logging into the system and reviewing the current stock levels, Monitoring and managing medicine stock involves tracking quantities, expiration dates, and essential details, with the system providing detailed medicine information. Admins can add, update, or delete stock, with the system reflecting changes. Generating comprehensive stock reports summarizes quantities, expiration dates, and key information, facilitating efficient stock management and ensuring availability.

Table 3. Use Case Description for Manage Stocks and Generate Inventory Reports

Use Case Name	Manage Stocks, Generate Inventory Reports
Participating Actors	Admin
Entry Conditions	• The admin initiates the process within the system. • The admin manage out of stock medicine and inventory management.
Flow of Events	1. The admin begins by monitoring the stock of medicines. 2. The system displays the details of the medicines to the admin. 3. The admin has the option to add stock. If yes, the admin inputs the details of the medicine to be added and displays the

	medicine details in the system section. If no, proceed to update the stock. 4. The admin has the option to update the stock. If yes, update the medicine details and display the medicine details in the system section. If no, delete the stock. 5. The admin has the option to delete the stock. If yes, delete the medicine details and display the medicine details in the system section. If no, end the transaction. 6. After displaying the medicine details in the system section, the admin has the option to generate a report. If yes, the system will generate reports. If no, it will end the transaction.
Exit Conditions	• The admin finishes overseeing the process, updates the inventory as needed, and makes sure the inventory reports in the system are accurate.
Quality Requirements	• The admin needs to keep the inventory reports accurate in the system by maintaining correct records of stock levels and transactions for effective stock control and inventory management.

Process Model: Activity Diagram

The flow of the activity diagram begins with the customer going to Arguelles Pharmacy. This streamlined activity diagram focuses on the interaction between the admin and the system, exhibiting the steps involved in encoding the customer's order, processing it within the system, and providing the necessary receipt to the consumer. This simple and clear illustration shows the critical processes in the order management process within the Medicine Inventory Management System.

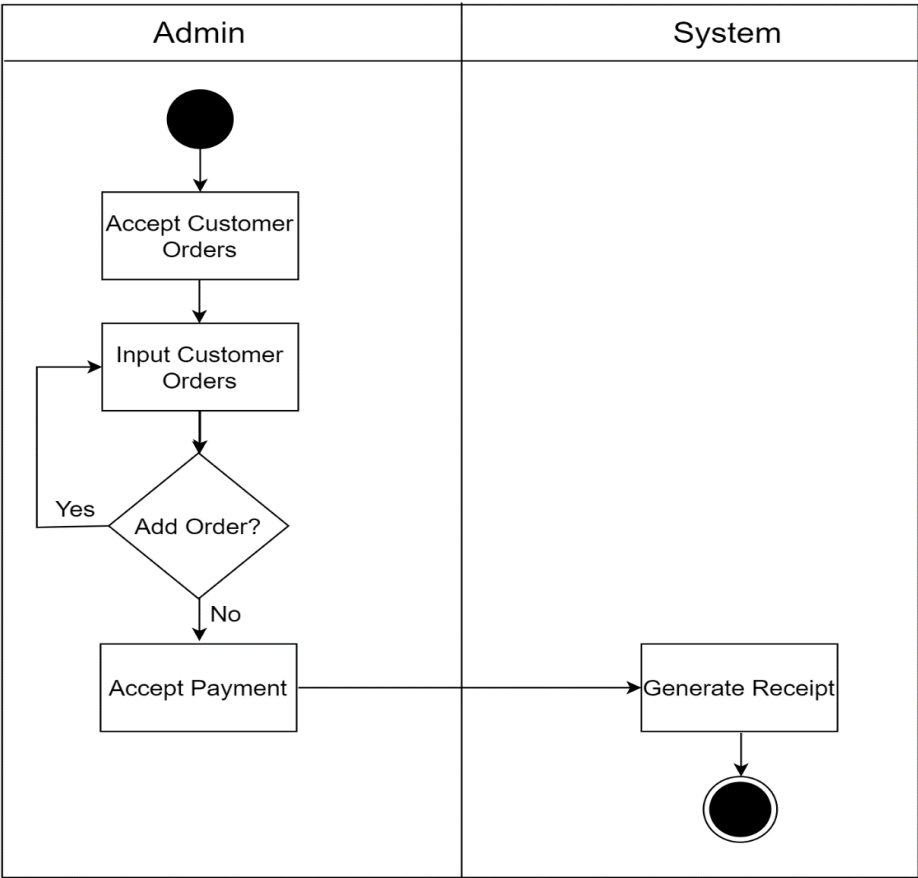


Figure 8. Activity Diagram for Manage Order and Generate Receipt

The visual representation of the process involved in managing medicine stock. It provides a clear and structured outline for an admin to follow. Starting from monitoring the stock, the admin can make decisions on adding new stock, updating existing stock details, deleting unnecessary entries, and generating reports. This flow enables efficient management of medicine stock within the system, ensuring that all necessary steps are taken to maintain accurate records and facilitate effective decision-making.

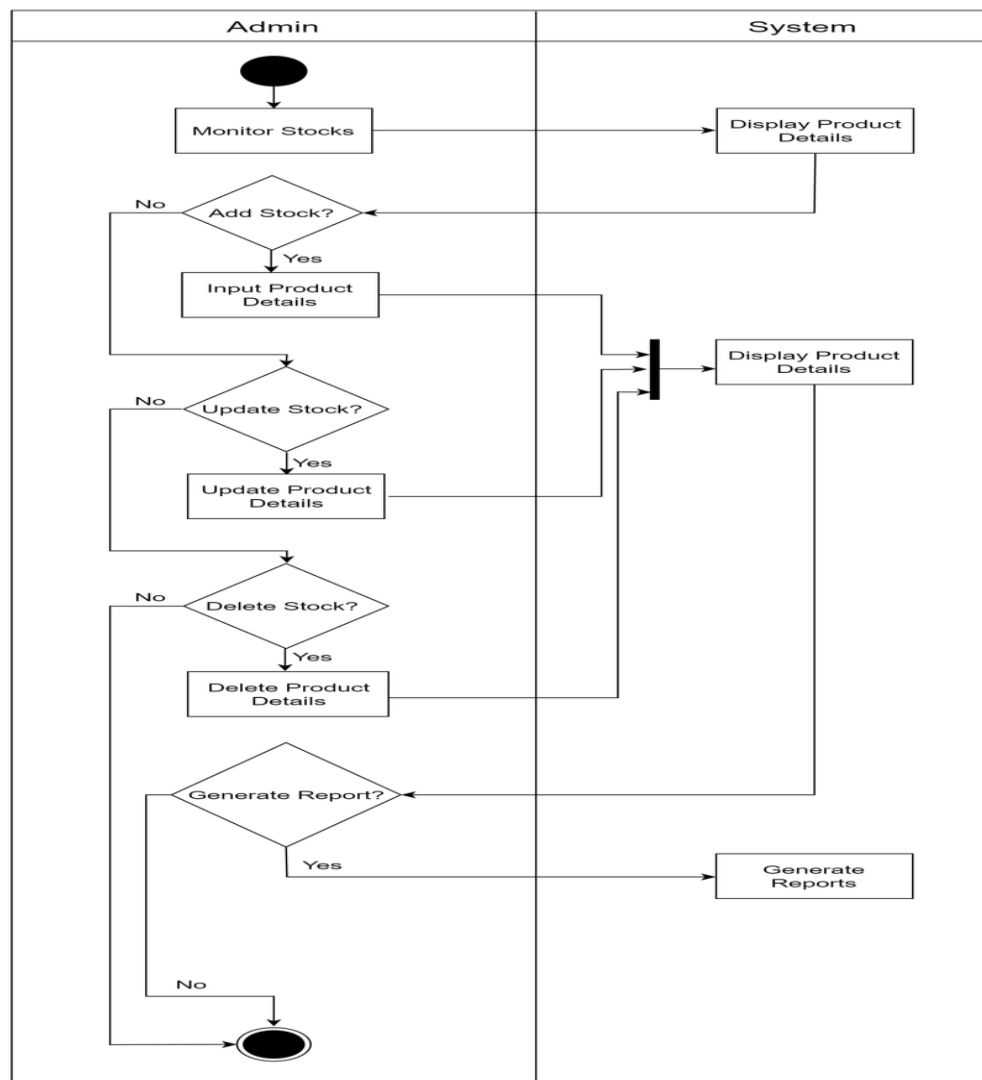


Figure 9. Activity Diagram for Managing Stocks and Generate Reports

Data Model: Context Data Flow Diagram

The Context data flow diagram is a high-level seeing representation showing the system's flow presenting a process, data accumulation and external operation with a system. It Consist of order details, receipt, medicine information, medicine details, inventory reports and sales report, depicting where data is stored [10]. The context date flow diagram helps the staff the overall data flow of the system without plunge into specified the details, it making it useful for the system and design accumulation.

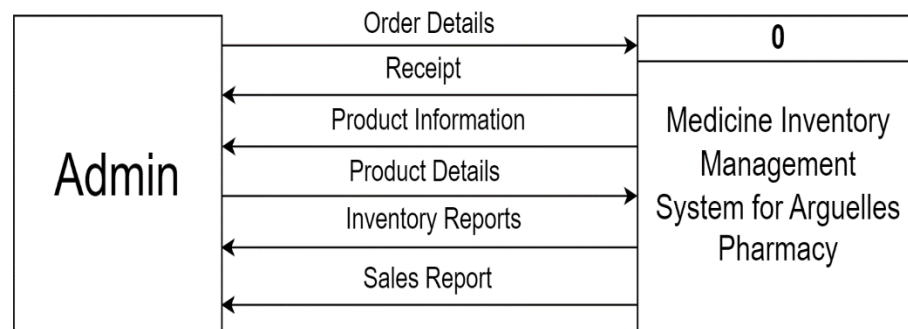


Figure 10. Context Data Flow Diagram

Data Model: Data Flow Diagram

The purpose of a level 1 DFD is to provide a clear and concise overview of the data flow within a system, allowing stakeholders to understand how information is processed and exchanged. It helps to identify the essential components of the system and their relationships, ensuring that all necessary inputs and outputs are accounted for [11]. The admin manages customer orders, enters them into an order data database, prints receipts for customers, and gathers sales reports. In addition, the admin monitors stock levels in the medicine data store and prepares inventory reports.

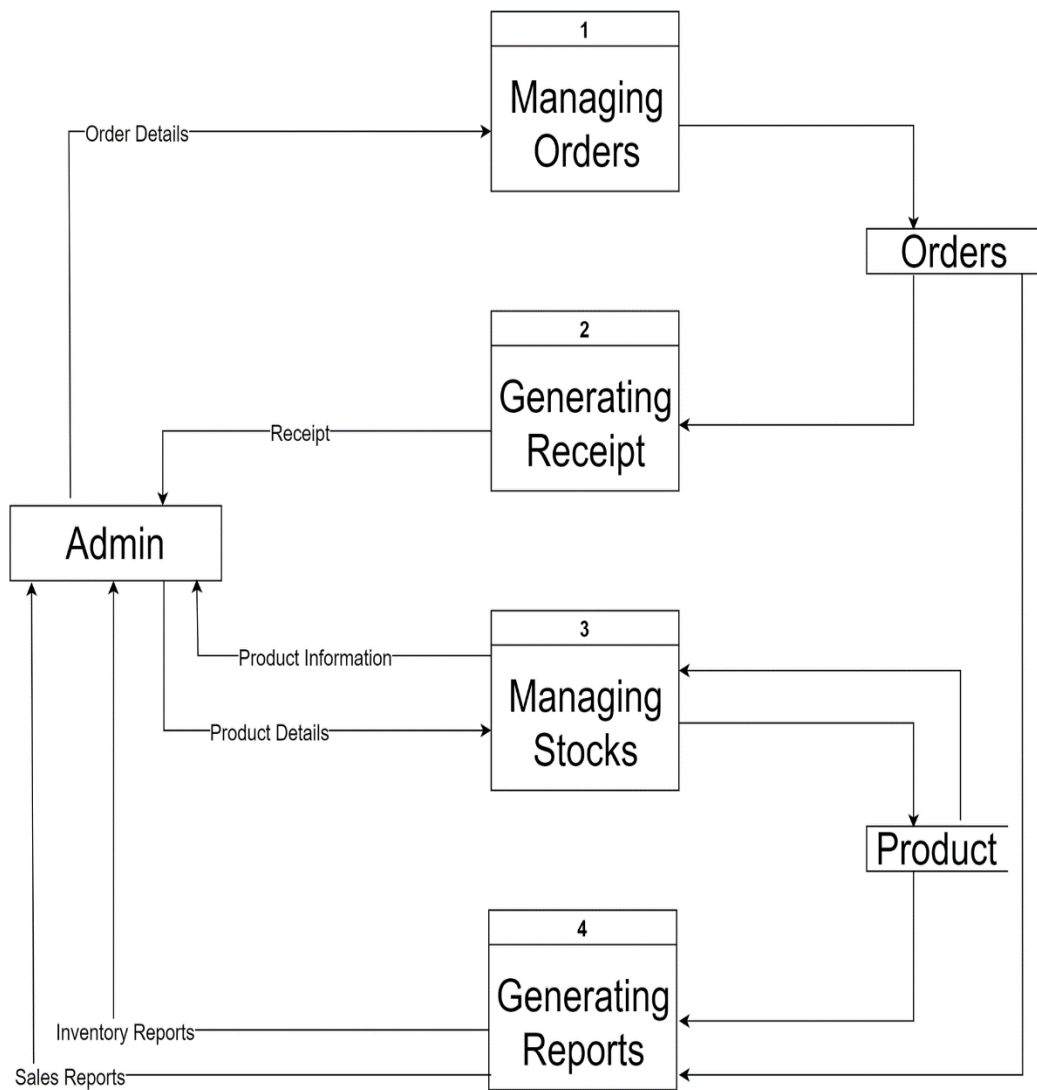


Figure 11. Level 1 Data Flow Diagram

Database Design: Entity-Relationship Diagram

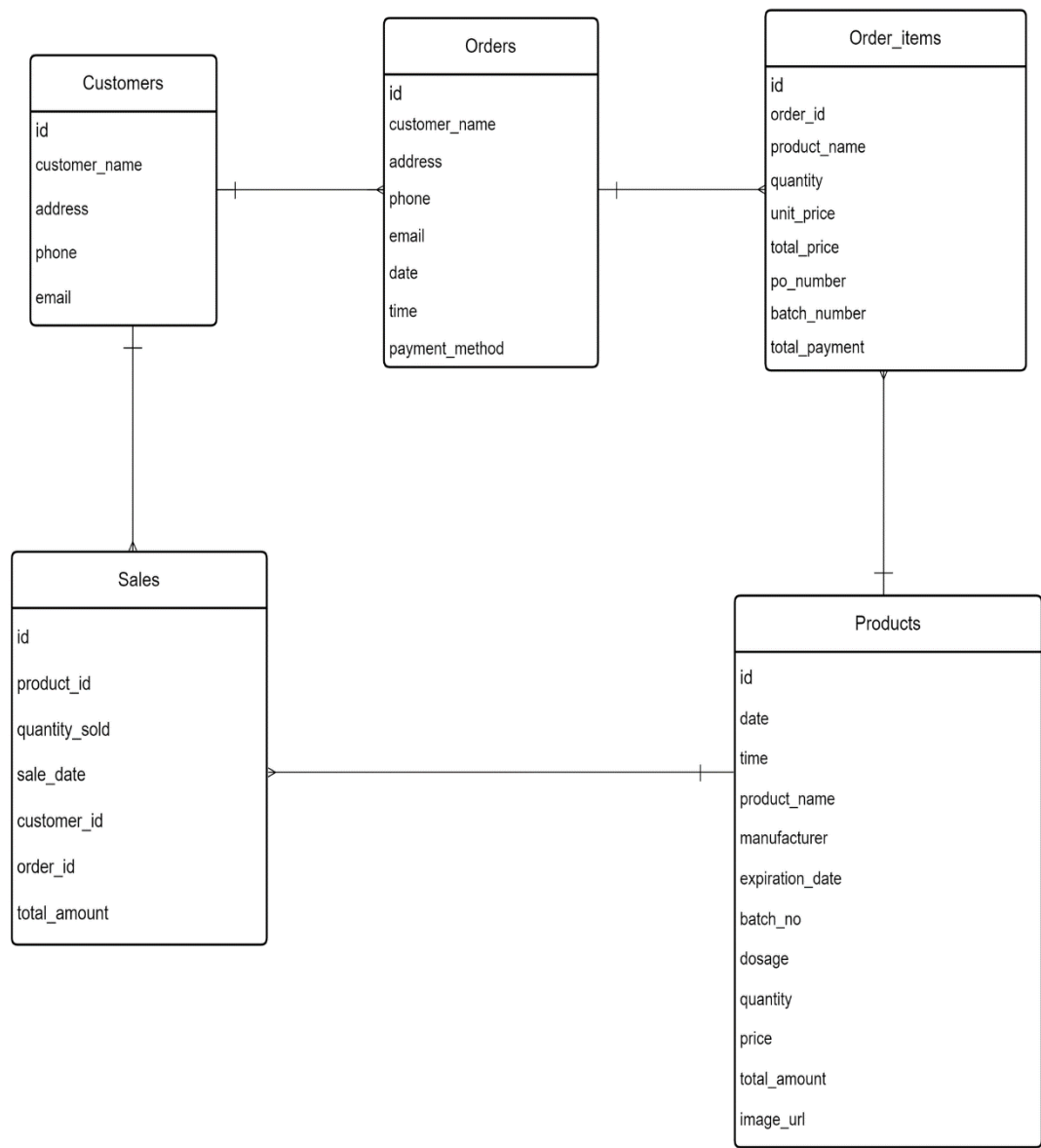


Figure 12. Entity-Relationship Diagram

CHAPTER III

RESULTS AND DISCUSSION

A. System Design

The system design focuses on enhancing business operations through a user-friendly platform that centralizes data and functionalities for managing customer interactions, inventory control, and sales analysis. Secure authentication protocols are integrated to ensure data security and user privacy, maintaining system integrity. This design adheres to industry best practices, aiming to optimize business operations and decision-making processes efficiently.

Data Dictionary

Customers

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
id	int	11	PK	Unique identifier for each record
customer_name	varchar	255	None	Name of a person or entity
address	varchar	255	None	Address of a person or entity
phone	varchar	50	None	Contact phone number
email	Varchar	255	UK	Email address (must be unique

Orders

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
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id	int	11	PK	Unique identifier for each record
customer_name	varchar	255	None	Name of a person or entity
address	varchar	255	None	Address of a person or entity
phone	varchar	50	None	Contact phone number
email	Varchar	255	UK	Email address (must be unique)
date	date	N/A	None	Date of transaction
time	time	N/A	None	Time of transaction
payment_method	varchar	50	None	Payment method used

Order_items

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
id	int	11	PK	Unique identifier for each record
order_id	int	11	None	Identifier for the order
product_name	varchar	255	None	Name of the product
quantity	int	11	None	Quantity of the product order

unit_price	decimal	(10;2)	None	Price per unit of the product
total_price	decinal	(10,2)	None	Price per unit of the product
po_number	varchar	50	None	Purchase order number
batch_number	varchar	50	None	Payment method used
total_payment	decimal	(10,2)	None	Total payment amount

Products

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
id	int	11	PK	Unique identifier for each record
date	date	N/A	None	Date of the record of transaction
time	time	N/A	None	Time of the record of transaction
product_name	varchar	255	None	Name of the product
manufacturer	varchar	255	None	Name of the manufacturer
expiration_date	date	N/A	None	Expiration date of the product
batch_no	varchar	50	None	Batch number of the product

dosage	varchar	50	None	Payment method used
quantity	varchar	50	None	Dosage information of the product
price	decimal	(10,2)	None	Price per unit of the product
total_amount	decimal	(10,2)	None	Total amount for the transaction
Image_url	varchar	255	None	URL of the product image

Sales

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
id	int	11	PK	Unique identifier for each sale record
product_id	int	11	FK	Identifier for the product
quantity_sold	int	11	None	Quantity of the product sold
sale_date	date	N/A	None	Date of sale
customer_id	int	11	FK	Identifier for the customer
order_id	int	11	FK	Identifier for the order

total_amount	decimal	(10,2)	None	Total amount for the sale
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System

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
username	varchar	50	None	Username of the person making the sale
password	varchar	255	None	Password of the person making the sale

User Interface Design

The log-in interface is the entry point for users to access the system. It typically includes fields for entering a username and password. The purpose is to authenticate users, ensuring that only authorized individuals can access the system's functionalities.



Figure 13. Log-in

The dashboard serves as the main interface after logging in. It provides an overview of the system's key metrics and quick access to various functionalities. This is where users can get a summary of important information and navigate to other sections.

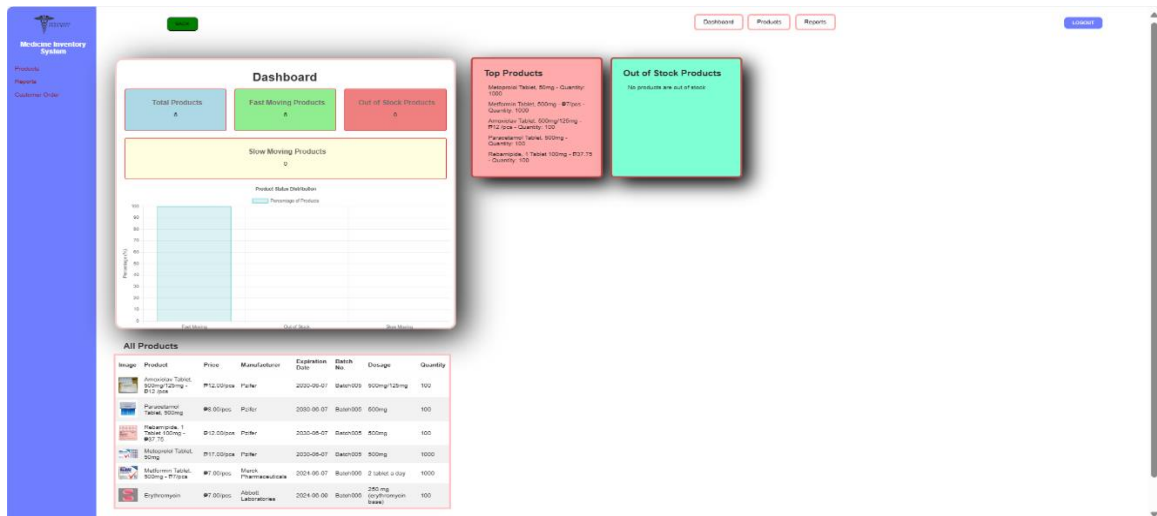


Figure 14. Dashboard

The product interface is where users can manage the inventory of products. Users can view, add, edit, and delete products. It ensures that the inventory is up-to-date and accurate.



Figure 15. Product

The order details interface displays all relevant information about a specific order. Users can view the order items, customer details, payment method, and total amount. It is essential for tracking and managing orders.

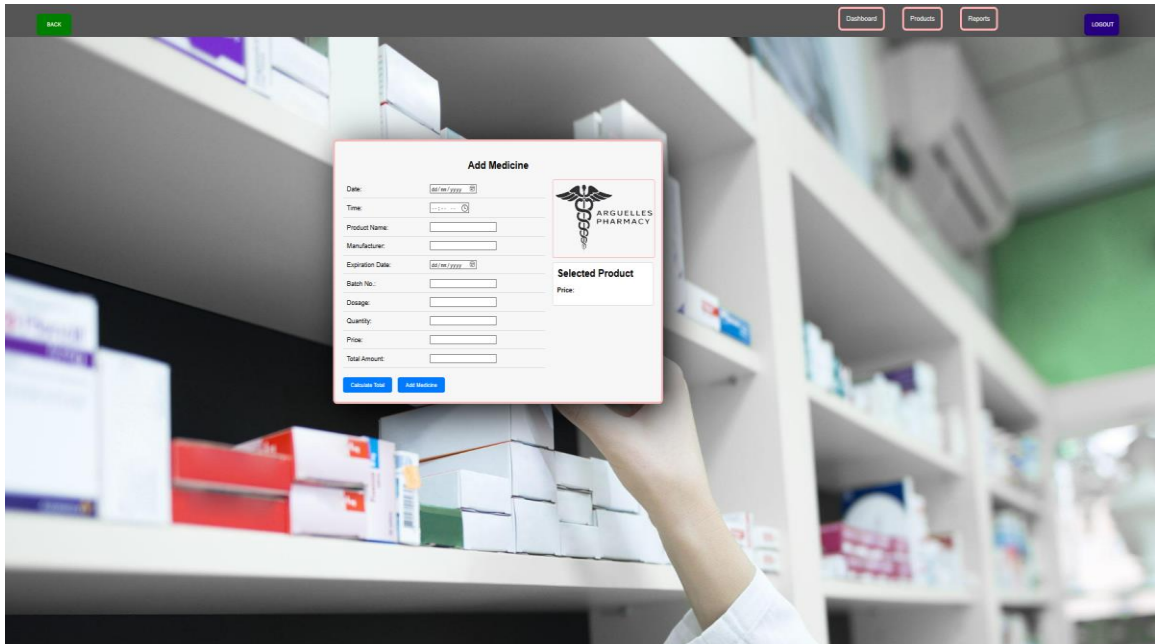


Figure 16. Order Details

Add medicine interface allows users to input new medicines into the system. It is crucial for maintaining an accurate and comprehensive inventory of available medicine.

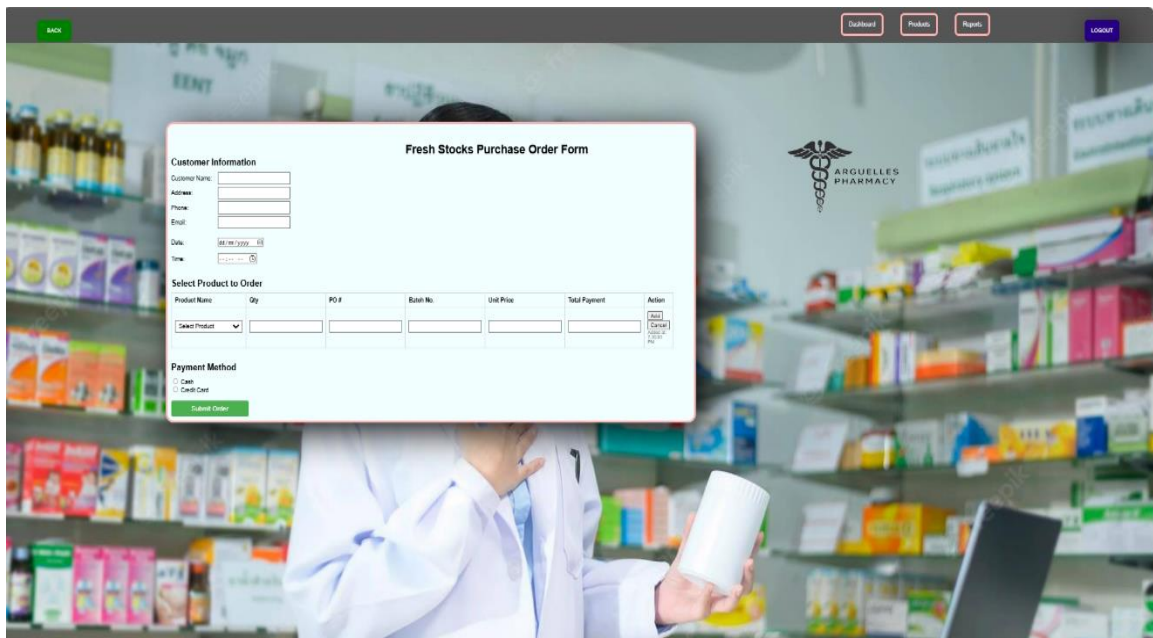


Figure 17. Add Medicine

Weekly sales report interface provides a detailed view of the sales data for a specific week. It helps users analyze sales performance over the past week, identify trends, and make informed business decisions. This report is crucial for monitoring short-term sales fluctuations and making timely adjustments.

Weekly Sales Report							
Sale Date	Customer Name	Order Date	Order Time	Product Name	Quantity	Unit Price	Total Amount
2024-06-09	allysa	2024-06-07	19:27:00	Amlodipine Tablet,200,Batch007	3	200.00	600.00
2024-06-09	Denver	2024-06-09	19:51:00	Aspirin Tablet,85,Batch004	3	85.00	255.00
2024-06-09	Arjay	2024-06-09	19:54:00	Loratadine Tablet,150,Batch005	3	150.00	450.00
2024-06-09	Jeannie Ann	2024-06-04	18:47:00	Cetirizine Tablet,50,Batch003	1	50.00	50.00
2024-06-09	Jhonry	2024-06-05	19:13:00	Aspirin Tablet,85,Batch004	3	85.00	455.00
2024-06-09	Jhonry	2024-06-05	19:13:00	Ibuprofen Tablet,100,Batch004	2	100.00	455.00
2024-06-09	Mela Aniza	2024-06-06	19:16:00	Paracetamol Tablet,120,Batch002	4	8.00	117.00
2024-06-09	Mela Aniza	2024-06-06	19:16:00	Aspirin Tablet,85,Batch004	1	85.00	117.00
Grand Total							2,499.00

Figure 18. Sales Report

The inventory report interface provides a comprehensive overview of the current stock levels of all products. It helps users manage inventory efficiently, ensuring that there is sufficient stock to meet demand and identifying products that need reordering.

Image	Product	Price	Manufacturer	Expiration Date	Batch No.	Dosage	Quantity
	Amoxiclav Tablet, 500mg/125mg - #12 pcs	฿12.00/pcs	Pzifer	2030-08-07	Batch005	500mg/125mg	100
	Paracetamol Tablet, 500mg	฿8.00/pcs	Pzifer	2030-08-07	Batch005	500mg	100
	Ribampide, 1 Tablet 100mg - #97.75	฿12.00/pcs	Pzifer	2030-08-07	Batch005	500mg	100
	Metoprolol Tablet, 50mg	฿17.00/pcs	Pzifer	2030-08-07	Batch005	500mg	1000
	Metformin Tablet, 500mg - #7/pcs	฿7.00/pcs	Merck Pharmaceuticals	2024-08-07	Batch006	2 tablet a day	1000
	Erythromycin 250 mg	฿7.00/pcs	Abbott Laboratories	2024-08-09	Batch006	250 mg (erythromycin base)	100
	Abacavir Sulfate 300 mg	฿15.00/pcs	Mylan Pharmaceuticals Inc.	2028-08-17	Batch008	300 mg	100
	Cephalexin Monohydrate 500 mg	฿17.00/pcs	Antipolo City	2028-12-02	batch 008	500 mg	200
	Cetirizine Hydrochloride 10 mg	฿4.00/pcs	Perigo Company	2028-03-21	Batch009	10 mg	200
	Loperamide Hydrochloride	฿8.00/pcs	Major Pharmaceuticals Inc.	2028-08-09	Batch009	2 mg	150
	Omeprazole Delayed-Release 40 mg.	฿15.00/pcs	Glenmark Pharmaceuticals Inc. USA	2028-11-13	Batch009	40 mg	150

Figure 19. Inventory Report

CHAPTER IV

CONCLUSION AND RECOMMENDATION

A. Conclusion

The Medicine Inventory Management System (MEDIMASYST) designed for Arguelles Pharmacy seeks to improve the pharmacy's operational efficiency and inventory control by facing major issues such as out-of-stock conditions, delayed deliveries, and reliance on manual record-keeping processes.

The system enables the administrator to effectively manage customer orders, track product details, and generate comprehensive reports, with features such as recording customer and order details, maintaining an up-to-date inventory database, automating order fulfillment and payment processes, and generating sales and inventory reports that support decision-making.

Arguelles Pharmacy should expect increased inventory visibility, fewer stockouts, and more effective operations after deploying this system, allowing the administrator to make educated decisions, ensure medicine availability, and give better customer care.

B. Recommendation

The findings indicate that MEDIMASYST has been beneficial in streamlining the day-to-day transactions at Arguelles Pharmacy. It is recommended that:

1. For students, it is important to enhance the data analytics capabilities of MEDIMASYST by focusing on developing advanced reporting and visualization tools. Additionally, incorporating predictive analytics can greatly improve decision-making processes.
2. Instructors should guide students in conducting thorough testing and validation of MEDIMASYST to ensure its reliability and adherence to industry standards. Encouraging students to explore integration possibilities with other healthcare systems will contribute to a more comprehensive solution.
3. Future researchers, implementing a notification system to address out-of-stock medicines in MEDIMASYST is crucial. This enhancement will greatly improve inventory management and ensure the availability of medicines for customers.

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APPENDICES

Letter to the Organization

**DAVAO DEL NORTE
STATE COLLEGE**
New Visayas, Panabo City, Davao del Norte, 8105

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Institute of Computing

January 30, 2024

MS. MAEBELLE FABAY
Owner
Arguelles Pharmacy
Somoso Anita Bldg.
Purok Mangga Brgy Sto Niño

Dear Ms. Fabay,

Warmest greetings!

We are second-year Davao del Norte State College (DNSC) students pursuing a Bachelor of Science in Information Systems this 2nd Semester of the School Year 2023-2024. As part of our course subject requirement for **IS221-System Analysis and Design**, we are asking your humble office to grant us permission to conduct our project.

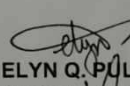
Relative to it, the undersigned are respectfully requesting in your good office permission to gather data by distributing online questionnaires/interviews and obtain an informed consent form of the participants involved in the project and will ensure that their privacy is protected.

Rest assured that the data gathered shall be used for the academic purpose of the study and will be treated confidentially. Your positive response regarding this request will significantly help the study's success.

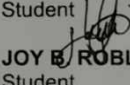
Thank you very much, and God bless.

Sincerely yours,

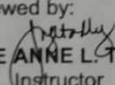

SHIENNA MAE D. DAKIT
Student

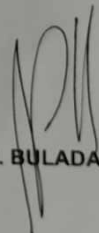

ELYN Q. PULGARINAS
Student


JEANNIE ANN A. RENTUAYA
Student

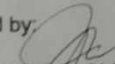

JOY B. ROBLEDO
Student

Reviewed by:

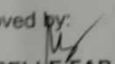

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IS221, Instructor


MARK VAN M. BULADACO, DBMIS
Dean, IC

Noted by:


JOVANNE C. ALEJANDRINO, MIS
Program Chairperson, BSIS

Approved by:


MAEBELLE FABAY
Owner, Arguelles Pharmacy

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Love of God and Country



Environmental Scanning Documentation



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Institute of Computing

February 21, 2024

MAEBELLE FABAY

Owner

Arguelles Pharmacy

Somoso Anita Bldg.

Purok Mangga Brgy. Sto Niño

Dear Ms. Fabay,

Warmest greetings!

With regard to our previous request to conduct an environmental scanning, we are pleased to inform you that our study entitled **Medicine Tracking System for Arguelles Pharmacy** has been accepted by the System Analysis and Design instructor as our final project.

Our final project requires us to plan, design, and create software to develop an IS solution. One of its goals is to computerize business processes that would address relevant societal problems as a requirement for completing our course.

Software development requires developing, gathering, analyzing, and testing our applications and data. Thus, we request your assistance in collecting data for our project for an accurate and reliable outcome. Further, we would like your expertise to help us continue realizing our software solutions through testing and interviews. Your expertise and knowledge in this area would be invaluable, and we would be very grateful if you could spare some time to help us.

We are hoping for your full support in the development of this project.

Thank you very much, and God bless.

Sincerely yours,

SHIENA MAE D. DAKIT
Student

ELYN Q. PULGARINAS
Student

JEANNIE ANN A. RENTUAYA
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